

PRAVEEN KUMAR

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 12+ years building and scaling high-traffic systems, including 5 years of technical leadership in product-based organizations. Deep hands-on expertise in PHP, Node.js, and Python, with a proven record of designing REST/GraphQL APIs, optimizing performance, and mentoring engineering teams for products serving millions of users. Transitioning into GenAI/AI Engineering with two applied, deployed agentic AI projects, and actively deepening skills in FastAPI, LangChain, LangGraph, MCP, CrewAI, and AI evaluation frameworks. Seeking GenAI Engineer, AI Engineer, or Senior AI Engineer roles that build on a strong backend and systems-design foundation.

TECHNICAL EXPERTISE

- **Backend:** PHP, Laravel, Node.js, Python, FastAPI
- **GenAI / Agentic AI (actively developing):** LangChain, LangGraph, LangSmith, MCP (Model Context Protocol), DeepAgents, CrewAI, Agentic AI design patterns, Google ADK, Portkey LLM Gateway, Guardrails AI, DeepEval, RAGAS
- **Architecture:** System Design, Scalable Systems, REST APIs, GraphQL, Microservices
- **Databases:** MySQL, MongoDB, Redis
- **Cloud & DevOps:** AWS, Docker, CI/CD
- **Leadership:** Technical Leadership, Mentoring, Code Reviews, Cross-functional Collaboration

TECHNICAL PROJECTS

Agentic RAG Assistant

GitHub: <https://github.com/praveenkumar/agentic-rag-assistant>

Built a FastAPI service orchestrated with LangGraph (branching retrieve-decide-tool-call-generate-guardrail workflow) backed by a locally persistent Chroma vector store, with an MCP stdio server exposing calculator and web-search tools for when local context is insufficient. Added LangSmith run tracing, inspectable guardrails (PII detection, heuristic groundedness, scope refusal), and an evaluation harness measuring retrieval hit@k and LLM-judged answer groundedness, deployed via Docker and Render with startup-time index rebuilding for ephemeral environments.

Tech: Python, FastAPI, LangGraph, LangChain, Chroma, MCP, LangSmith, Docker, pytest

Multi-Agent Task System, CrewAI

GitHub: <https://github.com/praveenkumar/multi-agent-task-system>

Built a CrewAI research-to-report pipeline with Researcher, Writer, and Reviewer agents; the Researcher uses live Tavily search, with the LLM provider configurable between OpenAI and Anthropic. Implemented a governed reject-and-revise workflow (maximum one revision) with structured handoff tracing, five evaluation criteria validated across five topics using deterministic fixtures, and a FastAPI /run endpoint returning the final report, evaluation scores, and full execution trace, deployed via a Render Blueprint with a /health check.

Tech: Python, CrewAI, FastAPI, Tavily API, OpenAI/Anthropic, Render

PROFESSIONAL EXPERIENCE

Senior Project Lead

Aug 2019 – Sep 2024

Matrimony.com Limited — Chennai, India

- Provided technical leadership for backend systems supporting millions of users, driving scalable architecture and engineering best practices.

- Led backend development while mentoring engineers through technical design discussions, code reviews, troubleshooting, and hands-on development.
- Collaborated with Product, QA, and Infrastructure teams to deliver scalable, reliable, and maintainable backend solutions.
- Participated in technical interviews and contributed to maintaining high engineering standards.

Key Engineering Contributions:

- Improved backend performance by approximately 30% through API optimization, MySQL query tuning, and Redis caching.
- Designed and developed scalable backend services for authentication and photo management modules supporting millions of users.
- Contributed to the migration of authentication and photo management modules from PHP to Node.js using GraphQL APIs.
- Developed reusable backend components to improve development efficiency and code consistency.
- Contributed to production stability through deployment support, incident troubleshooting, and performance optimization.
- Identified architectural bottlenecks and implemented scalable solutions to improve system reliability and maintainability.

Technologies: PHP, Laravel, Node.js, Python, GraphQL, REST APIs, MySQL, MongoDB, Redis, AWS, Microservices, Agile

Senior Web Developer

Jul 2018 – Aug 2019

Innoblitz Technologies — Chennai, India

- Delivered backend feature enhancements improving delivery efficiency by 20%
- Optimized database queries improving response performance by 40%
- Collaborated with product and QA teams for feature rollout and defect resolution

Lead Web Developer

Jun 2015 – Jul 2018

Times Partner Services Pvt Ltd — Chennai, India

- Led backend development for enterprise web applications
- Maintained 99.9% system uptime through performance and stability improvements
- Reduced development cycle time by 15% through reusable service components

Earlier Experience

Web Developer – Dot Infotech (Aug 2012 – May 2015)

Delivered web application modules, improved database performance, and supported full development lifecycle.

EDUCATION

Master of Science – Information Technology

University of Madras, Chennai

Bachelor of Computer Applications

D.G. Vaishnav College, Chennai

CERTIFICATIONS & CONTINUOUS LEARNING

- Generative AI (GUVI - HCL), Nov 2025
- AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents (Udemy), Sep 2025
- Machine Learning, Data Science & AI Engineering with Python (Udemy), Apr 2025
- AWS Certified Solutions Architect Associate (Udemy), Feb 2026 (Training & Preparation)
- Laravel - For Beginner to Advanced (Udemy), Nov 2024
- NodeJS - The Complete Guide (MVC, REST APIs, GraphQL, Deno) (Udemy), Nov 2024
- Building an AI ChatBot with OpenAI and Node.js (Tutorialspoint), Oct 2024